Ryan Sheatsley

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Academic & Professional Experience

United States Army Research Laboratory — **Adelphi, MD**
Research Scientist and Team Lead (mentored by Andrew Toth) — *Summers of 2014–2018*

Patent Litigation — **University Park, PA**
Code Review & Document Drafting — *2018–2019*

Systems Internet and Infrastructure Security Laboratory — **University Park, PA**
Systems Administrator — *2015–Present*

Penn State Learning Center — **University Park, PA**
Writing Tutor (mentored by Dr. Jon Olson) — *2012–2013*

Education

Pennsylvania State University — **University Park, PA**
M.S. in Computer Science and Engineering — *2016–2018*

- Thesis: *Adversarial Examples in Constrained Domains*
- Advisor: *Prof. Patrick McDaniel*
- Thesis committee: Prof. Patrick McDaniel, Prof. Trent Jaeger, Prof. Nicolas Papernot

Pennsylvania State University — **University Park, PA**
B.S. in Computer Engineering — *2011–2015*

Publications

Conference Proceedings

On the Robustness of Domain Constraints. *Ryan Sheatsley, Blaine Hoak, Eric Pauley, Yohan Beguin, Yohan Beugin, Michael Wiesman, Patrick McDaniel*. Proceedings of the 2021 Conference on Computer and Communications Security (CCS), Coex, Seoul, Republic of Korea. (2021)

Development of Machine Learning Algorithms for Directional Gamma Ray Detection. *Matthew Durbin, Ryan Sheatsley, Christopher Balbier, Tristan Grieve, Patrick McDaniel, Azaree Lintereur*. Proceedings of the 60th Annual Meeting of the Institute of Nuclear Materials Management (INMM), Palm Sptrings, CA. (2019)

Curie: Policy-based secure data exchange. *Z. Berkay Celik, Abbas Acar, Hidayet Aksu, Ryan Sheatsley, Patrick McDaniel, A Secuk Uluagac*. Proceedings of the Ninth ACM Conference on Data and Application Security and Privacy (CODASPY), Richardson, TX. (2019)

Network Traffic Obfuscation: An Adversarial Machine Learning Approach. *Gunjan Verma, Ertugrul Ciftcioglu, Ryan Sheatsley, Kevin Chain, Lisa Scott*. Proceedings of the 2018 Military Communications Conference (MILCOM), Baltimore, MD. (2018)

Detection under Privileged Information. *Z. Berkay Celik, Patrick McDaniel, Rauf Izmailov, Nico-*

las Papernot, Ryan Sheatsley, Raquel Alvarez, Ananthram Swami. Proceedings of the 2018 Asia Conference on Computer and Communications Security (ASIACCS), Incheon, Republic of Korea. (2018)

Heterogeneous information sharing of sensor information in contested environments. *Jason A Wampler, Chien Hsieh, Andrew Toth, Ryan Sheatsley*. Proceedings of Volume 10190 Ground/Air Multisensor Interoperability, Integration, and Networking for Persistent ISR VIII (SPIE), Anaheim, CA. (2017)

Journals

Physics-based Misbehavior Detection System for V2X Communications. *Alejandro Andrade Salazar, Ryan Sheatsley, Jonathan Petit, and Patrick McDaniel*. SAE International Journal of Connected and Automated Vehicles. (2021)

Improving Radioactive Material Localization by Leveraging Cyber-Security Model Optimizations. *Ryan Sheatsley, Matthew Durbin, Azaree Lintereur, and Patrick McDaniel*. IEEE Sensors. (2021)

Book Chapters

Evading Machine Learning-based Network Intrusion Detection Systems with GANs. *Bolor-Erdene Zolbayar, Ryan Sheatsley, Patrick McDaniel*. In: *Charles A Kamhoua, Christopher D. Kiekintveld, Fei Fang, Quanyan Zhu* (Ed.). Game Theory and Machine Learning for Cyber Security. John Wiley & Sons, Incorporated, Hoboken, New Jersey. (2021)

Technical Reports

A Vision Toward an Internet of Battlefield Things (IoBT): Autonomous Classifying Sensor Network. *John Zhu, Egan McClave, Quan Pham, Sujay Polineni, Sam Reinhart, Ryan Sheatsley, Andrew Toth*. (2018)

Cleverhans v2.0.0: an adversarial machine learning library. *Nicolas Papernot, Ian Goodfellow, Ryan Sheatsley, Reuben Feinman, Patrick McDaniel*. (2016)

Analyzing GAIAN Database (GaianDB) on a Tactical Network. *Ryan Sheatsley, Andrew Toth*. (2015)

Thesis

Adversarial Examples in Constrained Domains. *Ryan Sheatsley*. (2018)

Honors

Robert M. Owens Memorial Scholarship: Pennsylvania State University *2017*
Pathways Internship Program: United States Government *2015*
Peer Tutor Faculty Recommendation: Pennsylvania State University *2011*

Talks

Adversarial Examples in Constrained Domains: RIT Great Lakes Security Day *2020*
Adversarial Examples in Constrained Domains: Cyber-CRA Webinar *2020*

Professional Activities

Reviewer

TDSC: IEEE Transactions on Dependable and Secure Computing *2021*
IC: IEEE Internet Computing *2021*
TIP: IEEE Transactions on Image Processing *2020*
TEM: IEEE Transactions on Engineering Management *2020*
DKE: Data & Knowledge Engineering Journal *2020*
GLOBECOM: IEEE Global Communications Conference *2019*

Volunteer

Dancing with Robots Penn State Summer Camp Coordinator *2019*